

Sean Dickson

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PROFESSIONAL SUMMARY

Early career software engineer with production experience designing ETL pipelines, REST APIs, and AWS-backed data infrastructure in a biotech environment. Progressed from bench scientist to software engineer, delivering end-to-end solutions independently. Currently pursuing an M.S. in Artificial Intelligence at Johns Hopkins, with coursework in machine learning, AI-enabled systems, and algorithm design.

EDUCATION

Johns Hopkins University

M.S. Artificial Intelligence.

Expected: 2027

Relevant Coursework: Applied Machine Learning, Creating AI-Enabled Systems, Data Structures, Algorithms for Data Science, Linear Algebra, Probability & Statistics

University of Washington

B.S. Biochemistry.

Jun 2021

Relevant Coursework: Accelerated Intro to Computer Science, Physics, Calculus

EXPERIENCE

Consultant, Software and Data Engineering

Recombia Biosciences

Sep 2024 – Present

Remote

- Design and deploy Python-based ETL pipelines and bioinformatics scripts for automated data ingestion.
- Deploy AWS cloud services (EC2, S3, Lambda) to scale data processing workflows.
- Deploy REST APIs via GitLab CI/CD with structured validation, error handling, and automated health checks.

Research Associate I-II, Synthetic Biology Engineer

Recombia Biosciences

Sep 2021 – Sep 2024

San Francisco, CA

- Grew from Research Associate into the team's informal software engineer, owning end-to-end data infrastructure.
- Built REST API, SQL, and AWS integrations to load experimental data into the warehouse, replacing a manual workflow.
- Expanded tooling as new data needs surfaced, learning each layer of the stack on the job.

Intern, Computational Biology

University of California, Los Angeles

Jun 2020 – Jun 2021

Los Angeles, CA

- Built a Python tool that parsed metabolic pathway models from XML and simulated experiments across input combinations.
- Applied Shannon entropy to rank experiments by expected information gain.

Intern, Genetics Research

Harvard Medical School

Jul 2019 – Sep 2019

Boston, MA

- Wrote Python scripts to generate and evaluate novel biological sequences for a gene-editing research project.
- Built foundational skills in Python and Git through applied research work.

PROJECTS

Assay Data Uploader

- Built an automation tool that ingests raw lab instrument output, validates it, and registers results via authenticated REST APIs.
- Implemented structured error handling and logging for full traceability.

SecureBank

- Built and deployed an ML pipeline training multiple classifiers to identify fraudulent transactions.
- Deployed the model as a REST API with health checks and logging using Docker.

Huffman Encoder

- Implemented Huffman encoding for data compression using a min-heap priority queue built from character frequency analysis, recursion, and lossless encode/decode.

SKILLS

Programming: Python, Java, R, C, MATLAB, JSON, XML

Cloud & DevOps: AWS (Lambda, EC2, S3, DynamoDB), Docker, RESTful APIs, CI/CD Pipelines, GitLab

AI/ML: NumPy, Pandas, scikit-learn, PyTorch, OpenCV, RAG, Computer Vision, Dimensionality reduction, Classification, Regression

Backend: ETL Pipelines, SQL, Flask, Webhooks, pytest

Frontend: React, Typescript, HTML, CSS

Tools: Claude Code, Git, Jupyter, Benchling (API/SDK, Events, Insights, Workflows, Warehouse), Geneious, Linux, Visual Studio Code